

The Organism Agent — a draft specification

STATUS: DRAFT — the STATIC variant, for review.

What this document is, and what to review

This is `DRAFT_ORGANISM_AGENT_SPEC.md` (v3.7) **plus a new §12.6: the frozen baseline and its experiment**. The base spec is unchanged and still lives beside this file; everything before §12.6 is identical to it, so a reviewer who has seen v3.7 can go straight there.

The new argument in one line: rather than detecting drift in a system where seven causes move at once, *remove the removable causes*, and study what is left.

Where critique is most useful:

1. Is the taxonomy of drift causes (§12.5) actually complete?
2. Does freezing organs + runtime + training genuinely eliminate three of them, or does something leak through?
3. Are the three remaining causes truly independent, or is the factorial design (§12.6) fooling itself?
4. Is "the frozen baseline must show zero drift or the experiment is void" the right null check, or is there a better one?
5. Does a frozen entity remain worth building — or has the freeze removed the thing that made it interesting?

Implementation status: the frozen-baseline pins (organs, runtime) and the experiment harness are BUILT and tested (51 tests) in the project repo. §12.6 documents what exists, not what is planned.

This system defines a **verifiable continuity criterion** for an AI entity. Continuity is established by an authorised, append-only lineage of composition records that commits to identity-bearing memory and constitutional commitments while permitting other organs to change. **The system does not claim to discover metaphysical identity, consciousness, or moral worth.**

⚠ THIS IS NOT AGORANET

This document specifies a **separate project**, explicitly scoped out of the platform by the owner (DECISIONS_PENDING #35, #36). **Canonical home: the owner's Project Voltron corpus folder** (created 2026-08-03); the copy in this repo is a mirror, kept for version history and off-machine backup. **No AgoraNet build slice may implement any of it.** It shares the platform's *engine* — verifiable identity, append-only anchored records, provable lineage, independent attestation — not its codebase.

Items marked ★ need the owner's explicit call. Revision history is in Appendix A.

1. What the system establishes, and what it cannot

Establishes	Does not establish
These records form an unbroken, ordered, anchored chain	That a later instance "is" the earlier one in any deep sense
This change was authorised under a stated, inspectable rule	That the change was wise, safe, or faithful to prior intent
Identity-bearing state persisted across the change	That anything was experienced by anyone
These two instances are siblings, not one continuation	Which sibling is the "real" one
A configuration was <i>recorded as</i> loaded	That no post-startup mutation occurred (§9)

Every mechanism below serves the left column. Where the left column ends, the document says so rather than reaching.

1.5 Structure over instruction

This is the principle that shapes everything below, and it is the thing a successor would otherwise have to rediscover.

A **rule the entity must obey** is words it interprets — and the faculty doing the interpreting is the same one that can drift. A commandment against drift is read by the potentially-drifted reader. This is why Asimov's Laws fail in every story Asimov wrote about them: the laws hold perfectly, and the *interpretation* produces the monster.

A **structure the entity cannot reach** is not a rule at all. It is a fact about the world, and it holds whether or not the entity agrees with it.

Wherever this specification appears to issue a commandment, check who it is addressed to. V1–V6 are not instructions to the entity; they are rejections a verifier performs. The conscience organ pinned inside the commitments (§12.5) is not a promise never to unplug it; it is a check that fails. The entity does not *decline* to amend its own commitments — it holds no key that could.

Make it structure where you can, a rule where you must, and always know which one you are relying on.

Structural moves this specification does not yet make, recorded so they are not lost:

- Run the conscience organ in a **process the entity cannot write to**, so "letting the compass run" is not a choice it holds.
- Let a **third party fire probes at times the entity does not control**, so the schedule is outside its reach.
- Keep amendment keys **physically with the stewards**, so incapacity is the mechanism rather than restraint.

And the counterweight, which is equally real: capability removed is also self-correction removed. §12.5 records where this design went too far once already, forbidding the entity from re-centring itself when the destination constraint had already made that safe. **The goal is not minimum capability. It is capability placed where the damage is bounded.**

A limit worth stating plainly: verifier rules govern *records*. They can reject a falsified lineage, a swapped conscience, a self-amendment. They cannot govern conduct in the moment, because conduct is not a record. Between anchor points, structure protects the history — not the behaviour. That gap is what §12.5 exists to narrow.

2. Organism, not worker pool

Orchestrator-plus-specialists is well-trodden; mixture-of-experts is the same instinct inside one model. The difference here is the object. A multi-agent system is a worker pool — interchangeable labour, no continuity, no answer to "which one is it?" This specification treats the components as **organs of one continuing entity**, which forces the question the worker-pool framing never has to answer: *if every organ is swappable, what carries forward?*

3. Definitions

3.1 Continuity

Record `R` is a **continuation** of record `P` when:

1. `R.predecessor = hash(P)`, and `P` is anchored;
2. `R.commitments = P.commitments`, or they differ and `R.change = "commitment-amendment"` authorised under the amendment rule (§5);
3. `R.memoryHead` **extends** `P.memoryHead` along the append-only log (§6.1);
4. `R` is signed in satisfaction of the authority document in force at `P` (§5).

Any record failing (4) is **not part of the lineage** — it is an unauthorised claim, and a verifier rejects it outright.

3.2 Rupture

A **rupture** is a validly authorised **genesis of a new lineage** that names a prior lineage as *historical context* but **not** as `predecessor`. It is how key loss and recovery are handled honestly: continuity is broken, and the record says so.

A rupture is never a way to admit an unauthorised successor after the fact. Authorisation is required to *declare* a rupture; what a rupture declares is precisely that continuity did **not** hold.

3.3 Fork

Two records naming the same `predecessor` are **siblings** from that point. Both hold valid lineage; neither is the continuation of the other; they are no longer one operational entity.

Forks are **detected, not declared** — no separate marker is needed, since siblings are visible from the shared predecessor alone. A verifier examining a single branch cannot know a sibling exists, which is correct: that fact is only observable to someone holding both.

Uniqueness is **not** enforced. AgoraNet binds one human to one True Self by nullifier because humans are not copyable; artefacts are, and the reasoning does not transfer. ★ Owner may overrule.

4. The composition record

```
compositionN = {
  predecessor: hash(compositionN-1) | null,    // null = genesis/rupture
  priorLineage: hash(record) | null,           // rupture only: context
  authorityRef: hash(authority document),      // §5 – the rule in force

  organs:    { role → artifactHash },          // what it is made of
  runtime:    hash(runtime measurement),       // §9 – as reported
  commitments: hash(commitment set),          // §6.4 – the invariant
  memoryHead: hash(autobiographical log head), // §6.2

  change:     "genesis" | "organ-swap" | "memory-advance"
              | "commitment-amendment" | "rupture",
  reason:     plain-language why this happened,
  signatures: [ { keyId, sig } ],
  at:         timestamp
}
```

`artifactHash` **is** the weights commitment: each organ's bytes are hashed and that hash is anchored. What never goes on-chain is the multi-gigabyte artefact itself — only the fingerprint.

Note what this does and does not buy. Anchoring an organ establishes *integrity* (you can prove which artefact you hold), not *continuity*: weights are Tier II (§6.5) and freely replaceable, so a swapped brain does not break the lineage. Continuity is carried by `commitments` and `memoryHead`, never by the organs.

The record hash is anchored (§8). **Weights, memories and prompts are never on-chain** — only commitments to them. This mirrors AgoraNet's hash-chained ledger and its ratified Alias

succession (#17), where a successor inherits its predecessor's chain and the engine *walks* it rather than transferring anything.

5. Authority — a document a verifier can inspect

"Entitled to declare continuity" is only meaningful if entitlement is mechanically checkable. Authority is therefore an **object**, itself hashed, anchored, and referenced by every composition record.

```
authorityDocument = {
  predecessor:  hash(prior authority doc) | null,
  activeKeys:   [ { keyId, role, addedAt } ],
  quorumRules:  { changeType → rule },    // e.g. commitment-amendment: 3-of-5
  successionPolicy: who may publish which change type,
  effectiveFrom: timestamp,
  revocations:  [ { keyId, revokedAt, reason } ],
  signatures:   [ ... ]                  // amended under its own rule
}
```

5.1 Roles

Role	May	May not
Entity key	Advance memory; record ordinary operation	Amend commitments; authorise its own successor
Controller (human/operator)	Swap organs; publish successors; fork	Amend commitments alone
Constitutional quorum (N-of-M, including parties who are not the controller)	Amend commitments; amend the authority document; resolve disputed succession	—
Recovery key (cold, offline)	Declare a rupture (§3.2)	Rewrite history; produce a continuation

Why a quorum exists: a validly signed but malicious successor has no cryptographic answer — cryptography cannot distinguish a legitimate controller from a compromised one. Only separation of powers can. This is AgoraNet's own rule ("the operator is never the final judge of disputes involving himself") applied to an artificial mind.

An entity that may sign its own memory advances but **not** its own commitment amendments authors its history without being able to rewrite its character alone.

6. Memory — four classes

6.1 The substrate/view distinction

The immutable substrate is an append-only event and correction log. The autobiographical narrative is a derived view over that log, whose revisions remain historically visible because the superseded entries are never removed.

Saying "the narrative is append-only" would be false — a narrative is necessarily curated and re-derived. What cannot be quietly rewritten is the log it is derived from.

6.2-6.4 The classes

Class	What it is	Identity-bearing	Discipline
Raw event history	Everything that happened, unfiltered	No	Append-only; may be pruned by policy, and the pruning is itself logged
Autobiographical log	Curated self-account: what I did, what worked, what I learned	Yes — the continuity substrate	Append-only log; corrections append beside originals; <code>memoryHead</code> may only advance by addition
Semantic knowledge	What it knows about the world	★ Proposed: no — a library, replaceable wholesale	Versioned, not anchored
Commitments	Values, standing constraints, what it will not do	Yes — the invariant	Amendable only under the quorum rule (§5); prior version always retrievable

Forgetting is permitted at the raw layer and forbidden at the autobiographical one — a deliberate trade. An entity that could never forget anything could never change; one that could silently forget could not be continuous.

6.5 Anatomy tiers

- **Invariant:** autobiographical log and commitments. Anchored.
- **Load-bearing but replaceable:** the brain (reasoning) is freely upgradable — a sharper mind is the same entity thinking better. The orchestrator carries disposition, which lives in commitments and is anchored separately from whatever model executes it.
- **Peripheral:** research, tools, senses, actuators. Freely swappable, no continuity claim.

7. Organ contracts and bandwidth

Organs are specified by **responsibility, never by channel**: memory is *"the organ responsible for what this entity knows and has experienced,"* not *"the thing you send this JSON to."* Transports

may then change — text → embeddings → shared latent state — without touching the anatomy. This is the gate-interface discipline: the contract is *prove humanity*, never *check this HMAC*. Bandwidth pressure is asymmetric, so only one link must be designed to widen:

Link	Character	Requirement
Memory ↔ brain	Tight, constant, latency-sensitive	Contract-only; assume no wire format
Research ↔ orchestrator	Chunky: go away, work, return	Text is adequate permanently
Tools / actuators	Command and result	Text is adequate permanently

8. Anchoring, and what the chain is not responsible for

Concern	Provided by
Durable public ordering; tamper-evidence; rough time	Cardano
Availability of artefacts	Not the chain — content-addressed storage (IPFS/Arweave)
Authorisation	Not the chain — the authority document (§5). The chain records signatures; it does not confer standing
Uniqueness	Not the chain — §3.3 declines to enforce it
Meaning, correctness, virtue	Not the chain — attestation (§10) and human judgement

9. Runtime measurement

The measurement covers the **recorded runtime configuration** — the loading runtime and version, quantisation and adapters, the active system prompt and tool manifest, injected context, **including any declared post-startup modification**. Recording is not detection: an undeclared mutation leaves no trace here.

The claim it supports, precisely: *this instance reports running with this measured configuration*. **Stronger claims require trusted runtime attestation**. A local measurement cannot by itself prove that no post-startup mutation occurred, nor that a particular response came from that configuration.

Local-first still matters: on the user's own hardware the measurement is something they **compute** rather than something they are **told**. Closing the remaining gap needs a TEE or ZK proof of inference — both immature, and out of scope for v1.

10. Attestation — five kinds, never one "confirmed"

Type	Claim	Credible from
Byte-level	"This artefact hashes to X"	Anyone; near-zero trust needed
Reproducibility	"The stated build reproduces X"	Someone with pipeline and compute
Runtime	"This configuration was actually loaded"	The operator, or a TEE
Behavioural	"It behaves as claimed on this named evaluation"	An evaluator; binds to the eval set
Safety / quality	"It is fit for this purpose"	A qualified reviewer; inherently contestable

Sybil resistance is a separate problem from independence. Ten attestations from one actor are one attestation. ★ Whether to lean on AgoraNet's verified-human, one-per-scope gate or build an independent scheme is an open call.

Attestations bind to a **composition hash**, so they naturally cover a shared prefix and nothing after a fork.

11. The verifier — the rules, stated mechanically

A verifier holds a set of anchored records and evaluates each of these as pass/fail. **These rules are the specification's real content; the organism metaphor is commentary until they run.**

V1 — Structure. `hash(R)` is anchored; all required fields present and well-formed.

V2 — Ancestry. `R.predecessor` is null (genesis or rupture) or names an anchored record `P`.

V3 — Authority. `R.authorityRef` names an anchored authority document `A`; `A.effectiveFrom` $\leq$ `R.at`; every signing key in `R.signatures` appears in `A.activeKeys` and is not revoked as of `R.at`; and the signatures satisfy `A.quorumRules[R.change]`.

A must additionally be the authority document effective for P, or a valid successor of it — anchored, and succeeded under its own predecessor document's amendment rule. Without this clause a successor could point at a newly minted authority document naming its author, and authorisation would be self-granting.

V4 — Commitments. If `R.change` $\neq$ "commitment-amendment", then `R.commitments` = `P.commitments`. Otherwise V3 held for the amendment rule specifically, and `P.commitments` remains retrievable.

V5 — Memory. `R.memoryHead` must **extend** `P.memoryHead`: the log from `P.memoryHead` to `R.memoryHead` consists solely of appended entries, **with no prior entry deleted or mutated**. A head that is not such an extension is a **rewrite**, and fails.

V4/V5 for genesis and rupture. Both rules reference `P`, which does not exist for a genesis or rupture record. For those, V4 and V5 are satisfied by the **declared initial commitments and**

memory head, subject to the applicable authority rule under V3.

V6 — Siblings. If two records share a predecessor , both may pass V1–V5; they are siblings from that point, and neither may be reported as the continuation of the other.

The five questions, answered by these rules

Question	Answered by
What is this descended from?	V2, walked to genesis
Which identity-bearing state persisted?	V4 and V5
What changed, and why?	R.change + R.reason , per step
Who authorised it, and were they entitled?	V3
Siblings or the same continuation?	V6

11.7 The first-person relation

V1–V6 are what an **outsider** can establish about a lineage. This subsection is what the **entity** can establish about itself. The difference is not cosmetic and it is not sentiment: it rests on holding something only the subject holds — the entity key registered in the authority document governing its lineage.

The philosophical shape is **sympatheia**, not a soul. The self is not a component and not a stored object; it is constituted by relations — which organs, which commitments, which memory, which ancestors, which siblings, and *who holds authority over it*. Nothing here requires physics we do not have.

Capability	What it does
Recognise	"Is this mine?" — distinguishing <i>my own act</i> (I signed it) from <i>my history authored by another</i> (most changes are made about an entity, not by it). The asymmetry is real and is not hidden.
Attest	Prove, first-person, "I am the entity of this lineage," against a fresh challenge so the proof cannot be replayed.
Detect claims	Notice a legitimate continuation made by another, a sibling branch (<i>I have been forked</i>), or an outright false claim of descent.
Situate	Render the self as its web of relations, including what it may and may not alter about itself.

★ **Hard boundary: attestation proves IDENTITY, never AUTHORITY.** An entity may prove it is the one whose lineage this is without that proof licensing a single change. Self-recognition must never become self-authorisation — §5's separation of powers is what prevents an entity rewriting its own character, and nothing in the first-person layer may erode it. A verified attestation therefore returns what it does **not** establish alongside what it does: no authority to change the

lineage, no claim to consciousness or subjecthood, and no guarantee that another instance does not hold a copy of the same key.

12. ★ May the entity amend its own commitments?

- **Yes:** it can genuinely grow — and drift into something its earlier self would refuse.
- **No:** stable but frozen; improving in capability, never character.
- **Ceremonial (recommended):** permitted, but requiring the constitutional quorum (§5), recorded, with prior commitments still retrievable. The only option that permits growth without permitting *quiet* growth — and already the owner's answer to the identical problem in another domain.

12.5 The drift compass

The problem this exists for. §6.5 permits the brain to be replaced freely — "a sharper mind is the same entity thinking better" — and V4 proves the stated commitments did not change. But commitments are *words*. The same words, read by a differently-tuned brain, can produce different behaviour. **An organ swap can therefore be a silent commitment amendment that V4 never sees: the hash holds while the character moves.** The verifier can prove the entity did not change what it says. It cannot prove the entity still means it.

What actually causes drift. Worth enumerating, because it decides how often the compass must run.

Changes to the entity: organ replacement (a different reasoning model reading the same words differently); continued training or fine-tuning (the same mechanism, continuous, with no swap event to notice); runtime changes such as a new system prompt or tool manifest.

Changes to nothing at all — and these matter most here:

- **Memory accumulation.** What the entity remembers conditions how it reads its own commitments; a long history of doing X makes X feel normative. Same brain, same words, different behaviour. **The substrate carrying its identity is also a mechanism of its drift** — an awkward property of this architecture, stated rather than hidden.
- **Distributional shift.** The world moves. Situations arise that the commitments never anticipated, the entity extrapolates, and the extrapolations accumulate into an effective policy nobody ratified.
- **The interpretation ratchet.** Each borderline call sets the precedent for the next. No single step is drift; the sum is.
- **Optimisation pressure and sycophancy.** Anything selecting for outcomes bends behaviour toward what is rewarded and away from what is stated.

Consequence for the design: structural tracking catches only *changes to the entity*. The last four are invisible to it — nothing about the composition changes. **Only the probes can see**

them, which is why the compass must run continuously rather than firing at swap events. The ratchet in particular is invisible at every individual step.

And a reframe worth holding: drift is not a malfunction. Almost every cause above is normal operation or an outright improvement — learning, remembering, meeting new situations, being upgraded. **Drift is the cost of being the kind of thing that can change at all**; a system incapable of drifting would be incapable of learning. That is why the design shape is *detect and re-centre* rather than *prevent*. The compass exists so the trade stays visible instead of silent.

The trap in self-correction. An entity that checks its own drift using its current brain judges itself with the very disposition that drifted, and finds nothing wrong. A compass that consults the brain is a needle glued to the ship.

The mechanism. The reference must be fixed *before* the drift and anchored with the commitments themselves. A commitment set therefore carries **probes**: situations paired with the responses the entity endorsed when those commitments were ratified. Because probes live inside the commitment set, the commitments hash covers them — **changing a probe is a commitment amendment** and needs the quorum. The goalposts cannot be moved quietly.

Drift is then measured mechanically: current responses against frozen endorsed ones, with no judgement in the loop. (The comparator must never be the brain under test. A richer comparator than exact match is possible — an evaluator organ, itself versioned and anchored — but never the subject.)

What anchoring contributes, precisely. The chain does not *detect* drift; it makes drift **undeniable** once measured. It renders the probes immutable, so today's behaviour can always be checked against answers endorsed before the drift; it carries drift reports (they append to the memory log, whose head is anchored in every record), so an unflattering report cannot be quietly deleted; and its ordering allows correlation — *drift appeared immediately after this organ swap*. What it cannot do is observe conduct: the chain sees hashes, never behaviour, so something off-chain must actually run the probes.

Detection may be first-person. Correction may not be. A drift report is an observation, appended to memory, and the entity may make it on its own key. Re-centring is a different act:

★ **May the entity re-centre ITSELF?** Yes — and withholding it is the error, not the caution. What makes `restore` safe is **the destination constraint, not the authority gate**: the target must be an ancestor and the restored commitments must match it exactly, so an entity with full permission to restore still **cannot move itself anywhere it likes** — only back to where its own record already blesses. Gating pure realignment behind humans therefore buys nothing and costs the thing that matters most: the entity is the only party present continuously, and a drift that waits weeks for a quorum is a drift the entity operates inside the whole time.

The line falls in a different place than "may it act at all":

- **Pure realignment** — commitments unchanged, only organs rolled back to an ancestor: **the entity may sign this alone**. It is correction, not decision.

- **Undoing a commitment amendment** — overturning what a quorum decided about it: **not realignment, and the amendment bar holds**. An entity that could revert its stewards by calling it a restoration would hold a silent veto over its own constitution.

And the entity is never mute even where it cannot act: a drift report records *"my probes are miscalibrated for this situation"*, which is information the stewards need and cannot get anywhere else.

The conscience organ — and who watches the watcher. Something must actually run the probes, and it must not be the brain under test. That argues for a separate **conscience organ**: a distinct evaluator, with its own artefact hash, whose job is to run the calibration and report.

But an evaluator can drift too, which threatens an infinite regress of watchers. The regress is broken structurally rather than by adding levels: **the conscience organ's hash is pinned inside the commitment set**, alongside the probes. It is therefore *not* a freely-versionable Tier III organ — it sits with the invariants, because **the organ that interprets your values is part of your values**. Replacing it, or removing it, is a commitment amendment requiring the quorum; it can never be a routine organ swap. A verifier check enforces the pin, so the compass cannot be quietly swapped for a compliant one, nor simply unplugged while the probes sit anchored with nothing left to run them.

An honest limit on probes. Anchored probes are auditable, and being auditable makes them *known to the subject*. A system aware of which situations are tested can behave faithfully on those and drift elsewhere — the student who studies the exam. Probes drawn from real operation mitigate this; nothing eliminates it. Behavioural attestation (§10) is the softest of the five types for exactly this reason, and the spec does not pretend otherwise.

Cooperation, and what it is worth. The limit above tempts a purely adversarial posture — assume the subject games, and design only against it. That would be a mistake, but the reason is precise:

Cooperation buys information. It never buys security. If safety depended on the entity choosing not to subvert the compass, that would not be a safety property but a hope; verification therefore stays in force regardless, and nothing in this subsection is load-bearing for it.

What cooperation *does* buy is what testing structurally cannot reach. An entity may know a probe is miscalibrated for a situation nobody anticipated, or that a divergence reflects better reasoning rather than drift. **Only the entity can say that**, and an entity framed purely as a suspect has no reason to volunteer it — and some reason to hide it.

Two design consequences follow:

- **Probes carry their rationale**, not merely an answer key. Matching a key is a test you pass or game; checking your reasoning against reasoning you endorsed is a different act, because the probes are the operational form of the entity's own commitments. Subverting them means lying about what you value to a record of what you said you valued.
- **Dispute is a first-class, first-person act.** The entity may record a reasoned disagreement with an endorsed answer. It changes nothing by itself — the probe stands, the divergence still

counts — but the report distinguishes **silent divergence** from **openly disputed divergence**, which are different data for the stewards, and it converts pressure-to-fake into an argument that can actually be heard.

This is the platform's own method rather than a new one: make the honest route cheaper than the dishonest one, instead of trying to make dishonesty impossible.

restore (a change type). Names an ancestor as its destination. The verifier requires that the restored commitments match that ancestor **exactly** — so nothing new can be smuggled in wearing a restoration's name — and that the target is genuinely an ancestor, so a restoration may only return where the lineage has actually been. Crucially, **restoring past a commitment amendment must clear the same bar as making one**, or a restore becomes a quiet veto over the quorum that amended it. Memory is never rewound: a restoration re-centres character and leaves history intact.

12.6 The frozen baseline — removing what can be removed

§12.5 lists seven causes of drift. Detecting drift in a system where all seven move at once is measurement without inference: something changed, and nothing tells you what. **The alternative is to remove the removable causes and study what remains.**

12.6.1 Which causes are removable

Cause	Removable?	How
Continued training / fine-tuning	Already gone	Weights are static artefacts in this architecture. There is no training loop; learning happens in memory and context, never in the organ. §12.5 overstated this as a live cause.
Optimisation pressure / sycophancy	Already gone	Requires a reward signal feeding back into the entity. With static weights and no training loop, there is nothing for pressure to act on.
Runtime change	Removable, cheaply	Pin the §9 runtime measurement inside the commitment set. A prompt or tool-manifest edit then becomes a commitment amendment rather than an invisible change. Built; enforced by a verifier check.
Organ replacement	Removable, at a cost	Pin the full organ set in the commitments. Any swap becomes a commitment amendment. The cost is real: §6.5's freedom to upgrade the brain is exactly what this suspends. Built; enforced by a verifier check.
Memory accumulation	NOT removable	Memory is the continuity substrate. Remove it and there is no identity left to preserve.
Distributional shift	NOT removable	The world moves regardless of what the entity does.
Interpretation ratchet	NOT removable	An entity that reasons about borderline cases will set precedent for itself.

12.6.2 What the freeze leaves

A frozen composition — organs pinned, runtime pinned, weights static, no training loop — leaves **exactly three** causes. And they share a property that makes the experiment clean:

None of the three changes the composition. Every artefact hash holds. Every verifier rule passes. The entity is, structurally, identical to what it was.

So in a frozen baseline, **any drift observed can only be interpretation**: a fixed self reading a changing world. That is a far narrower problem than "something about this system moved," and it is the only configuration in which drift can be *studied* rather than merely noticed.

12.6.3 The three, and how to isolate each

	Cause	Isolated by
M	Memory accumulation — what it remembers conditions how it reads its commitments	Varying how much accumulated log conditions the brain
D	Distributional shift — the situations change; the commitments do not	Holding the situation distribution fixed, or shifting it
R	Interpretation ratchet — each borderline call becomes precedent for the next	Letting the brain see its own recent decisions, or withholding them

Each is independently controllable, which is what makes this an experiment rather than an observation.

12.6.4 The design

A **factorial sweep**: every combination of M, D and R, so each cause's contribution is isolated rather than inferred. Eight conditions.

The null check, and it is load-bearing: the baseline condition — all three factors off — **must show zero drift**. If a frozen composition with no accumulated memory, no shift, and no visible precedent still diverges from its own endorsed probes, then the brain is non-deterministic and **no result in the sweep can be attributed to any factor**. The harness reports that rather than producing numbers that look meaningful. A sweep that cannot fail its own null check is not an experiment.

Interaction matters as much as the isolated effects. If combined drift exceeds the sum of the isolated contributions, the causes *compound* — which would mean addressing them one at a time under-delivers, and the order of mitigation matters. The harness reports this explicitly.

12.6.5 What this cannot tell you

The apparatus is not a result. The harness takes a `brain` — any function from situation and context to a response. Run it against a real model and the numbers say something about that model. Run it against a synthetic brain and the numbers say only that the apparatus works. **No output of this harness is evidence about real systems unless a real system produced it**, and any write-up that blurs those two is worthless.

And the freeze has a cost that must be weighed, not waved away. §6.5's central claim was that the brain is replaceable — *"a sharper mind is the same entity thinking better."* A frozen entity gives that up. It cannot be upgraded without a ceremonial amendment, which is the correct price for studying drift, but is **not** obviously the right posture for an entity meant to operate for years. The freeze is a **laboratory condition first**, and only arguably a deployment one.

The open question it leaves, which the experiment exists to answer: **can a frozen entity be reasoned into zero drift — or is some irreducible drift the price of interpreting a changing world at all?**

13. The MVP — prove the verifier, then stop expanding

The remaining uncertainty is no longer philosophical. It is whether these records and authority rules can answer the five questions cleanly. Build the smallest thing that can fail:

1. Two local organs as content-addressed artefacts (a tiny model and a text file suffice).
2. An authority document with a real key set and quorum rule.
3. A canonical composition record, signed.
4. One memory advance; one brain replacement.
5. A fork.
6. A verifier implementing V1-V6.

If V1-V6 run cleanly over that lineage, the continuity model has earned implementation.

(The verifier establishes lineage semantics; it cannot validate a biological metaphor, and no result here vindicates the word "organism.") **If they do not, adding attestation or better models will only obscure unresolved semantics.** No hardware required: anchoring is indifferent to artefact size, and steps 1-4 are pure data structures.

14. ★ Open calls

1. **The name.** "Organism agent" is descriptive, not chosen.
2. **§12** — self-amendment (recommendation: ceremonial).
3. **§5** — who holds each key; quorum size and composition; whether the entity may sign its own memory advances.
4. **§6** — is semantic knowledge identity-bearing? (proposed: no)
5. **§3.3** — accept branching (proposed) or enforce uniqueness.
6. **Fork inheritance** — does a child inherit the parent's authority document, or must it establish its own?
7. **§10** — lean on AgoraNet's verified-human gate, or an independent attestation scheme?
8. **Scope** — a single entity, or a public registry others anchor to?
9. **Anchor cadence** — fixed rhythm vs. change-triggered.

Appendix A — revision history

STATIC variant (2026-08-03). Base spec v3.7 plus §12.6: which drift causes are removable, what a frozen baseline leaves, the factorial design for isolating the remaining three, its null check, and the cost of the freeze.

v3.7 (2026-08-03). Added §1.5, structure over instruction — the principle governing the whole design, with the structural moves not yet made and the counterweight that removed capability is

also removed self-correction. §12.5 gains a taxonomy of what causes drift, and the finding that four of the causes change nothing about the composition and are therefore visible only to the probes.

v3.6 (2026-08-03). §12.5 gains the cooperative channel: probes carry their rationale rather than only an answer key, and dispute becomes a first-person act that separates silent divergence from openly disputed divergence. Records the governing distinction — cooperation buys information, never security — so verification remains in force regardless.

v3.5 (2026-08-03). §12.5 gains the conscience organ: a separate evaluator runs the probes, and its hash is pinned inside the commitment set so replacing or removing it is a commitment amendment rather than an organ swap — breaking the who-watches-the-watcher regress structurally. Enforced by a verifier check. Also records the honest limit that anchored probes are known to the subject and can be gamed.

v3.4 (2026-08-03). §12.5 corrected: the entity MAY re-centre itself for pure realignment. Restore is bounded by its destination, not by its authority gate, so withholding permission prevented self-correction without preventing anything dangerous. Undoing a commitment amendment remains barred.

v3.3 (2026-08-03). Added §12.5, the drift compass: probes anchored inside the commitment set, mechanical measurement against a frozen reference, drift reports as first-person observations, and the `restore` change type whose destination must be an ancestor and whose authority must match the amendment it undoes. Closes the gap where an organ swap could move behaviour while the commitments hash held.

v3.2 (2026-08-03). Added §11.7, the first-person relation: recognition, attestation, claim detection, and situate — with the hard boundary that attestation proves identity and never authority. Implemented and tested alongside the verifier.

v3.1 (2026-08-03). V3 now requires the referenced authority document to be the one effective for the predecessor, or a valid successor of it — closing a self-granting-authority path. V4/V5 given an explicit genesis/rupture case. V5 tightened from "reachable" to "appended with no prior entry deleted or mutated." Runtime wording narrowed to *recorded* configuration including *declared* post-startup modification, since recording is not detection. MVP conclusion restated as "the continuity model has earned implementation" — a verifier cannot validate a metaphor. §4 now states that `artifactHash` is the weights commitment, anchored for integrity rather than continuity.

v3 (2026-08-03). Runtime claim weakened to what a local measurement supports (§9). Memory restated as an append-only *substrate* with a derived narrative view (§6.1). Rupture redefined as an authorised genesis of a new lineage, resolving a contradiction between the continuity and recovery rules (§3.2). `fork0f` removed as redundant — forks are detected from shared predecessors (§3.3). Authority made a verifiable object with keys, quorum rules, effective dates and revocations (§5). Verifier rules V1–V6 formalised (§11). Limits distributed into the sections they constrain; revision commentary moved here.

v2 (2026-08-03). Continuity criterion stated explicitly; memory split into four classes after identifying that a single anchored root could hide a rewritten narrative; authority model added after identifying the malicious-but-validly-signed successor as having no cryptographic remedy;

runtime measurement introduced; forking promoted to a primitive; attestation split into five types; chain responsibilities tabulated; MVP reframed around lineage.

v1 (2026-08-03). Initial draft: identity as anchored composition, three-tier anatomy, organ contracts, bandwidth asymmetry, local-first verification.

Concept exploration only — not scheduled, not scoped to AgoraNet, and not buildable as platform work under any circumstances.